

ORSK, Russian Federation

WMO: 351380

Lat: 51.067N

Lon: 58.600E

Elev: 935

StdP: 14.21

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-15.0	-25.8	1.4	-18.8	-21.7	1.7	-15.1	31.9	23.5	28.6	22.6	7.6	30	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	94.4	64.8	91.2	64.1	87.9	63.6	68.4	85.5	66.9	83.9	65.3	82.5	12.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	88.4	73.4	60.9	82.6	72.7	59.0	77.3	72.4	33.2	86.1	31.8	82.9	30.7	81.9	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.1	21.4	DB	-23.9	99.3	5.4	2.8	-27.8	101.3	-31.0	102.9	-34.1	104.5	-38.0	106.5	(4)
(5)			WB	-24.3	71.6	5.3	1.8	-28.1	72.9	-31.2	73.9	-34.2	74.9	-38.1	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	4.7	6.6	22.8	45.1	59.9	68.9	72.1	70.5	58.0	42.7	26.9	13.0		
	DBStd	26.21	12.42	11.74	10.27	10.74	8.46	7.93	6.53	7.98	8.52	9.07	11.47	12.08		
	HDD50	5813	1403	1214	844	215	15	1	0	0	24	259	692	1146		
	HDD65	9404	1868	1634	1309	597	205	50	17	40	241	690	1142	1611		
	CDD50	2574	0	0	0	69	322	568	684	634	263	34	0	0		
	CDD65	690	0	0	0	1	47	167	236	209	30	0	0	0		
	CDH74	8181	0	0	0	41	605	2004	2684	2489	354	4	0	0		
	CDH80	3460	0	0	0	4	162	861	1159	1155	119	0	0	0		
(14) Wind	WSAvg	10.8	10.7	10.9	12.3	11.8	11.1	10.2	10.1	9.8	10.1	10.9	11.2	10.6		
(15) Precipitation	PrecAvg	13.5	1.0	0.7	1.1	1.1	1.3	1.5	1.7	1.3	0.7	1.0	0.9	1.1		
	PrecMax	19.9	1.7	1.6	2.6	3.0	2.9	3.8	3.2	3.5	1.8	1.9	2.0	2.2		
	PrecMin	8.2	0.2	0.1	0.0	0.3	0.2	0.5	0.4	0.2	0.2	0.0	0.0	0.3		
	PrecStd	3.3	0.4	0.4	0.7	0.7	0.7	0.8	1.0	1.0	0.6	0.5	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	34.6	53.8	79.2	87.8	96.6	98.6	98.8	91.3	73.1	51.7	37.5		
		MCWB	34.1	33.7	44.7	56.4	62.1	64.9	65.0	65.0	61.0	52.9	45.0	34.2		
	2%	DB	30.6	32.1	42.7	73.0	84.3	92.8	93.4	93.6	82.5	66.4	47.8	34.0		
		MCWB	29.7	31.2	37.5	54.2	60.5	64.2	65.0	64.6	57.7	49.5	43.1	32.7		
	5%	DB	28.2	28.8	37.4	67.8	80.9	89.4	89.8	91.0	77.3	61.1	44.2	32.1		
		MCWB	27.2	27.8	34.7	52.5	59.8	63.2	64.6	64.5	57.1	48.4	41.2	30.9		
	10%	DB	23.4	24.9	35.2	62.5	76.9	84.7	86.3	86.4	73.2	55.8	40.4	29.9		
		MCWB	22.4	24.0	33.2	49.9	58.5	62.6	63.8	63.1	55.2	46.6	38.0	28.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	33.8	44.9	59.4	66.6	70.1	70.4	70.8	62.8	56.4	47.3	35.5		
		MCDB	34.2	34.4	52.8	74.5	81.4	87.6	85.9	89.3	80.9	65.8	49.3	36.1		
	2%	WB	29.7	31.0	37.8	55.9	63.6	67.7	68.1	68.1	60.2	52.4	43.7	32.8		
		MCDB	30.3	31.7	41.4	69.6	78.7	84.7	84.4	86.8	76.9	61.5	45.8	33.5		
	5%	WB	26.7	27.9	34.9	53.3	61.4	65.7	66.8	66.0	58.3	49.7	40.5	30.6		
		MCDB	27.7	28.8	36.5	65.6	76.4	82.5	83.4	83.9	74.1	58.8	43.2	31.6		
	10%	WB	22.5	23.9	32.9	50.5	59.6	64.0	65.3	64.4	56.4	47.3	37.6	28.2		
		MCDB	23.4	24.8	34.3	60.9	74.0	80.4	82.2	83.2	70.8	55.0	39.9	29.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	13.9	14.0	19.8	23.2	23.5	23.1	23.7	21.7	17.2	11.5	12.3		
		MCDBR	10.5	11.9	15.5	27.2	27.4	28.4	27.2	28.0	26.7	25.6	15.4	10.8		
		MCWBR	10.5	11.5	12.1	14.1	11.9	11.2	10.3	10.9	11.7	13.6	11.6	10.1		
	5% WB	MCDBR	11.1	11.6	14.2	25.3	24.8	24.8	24.2	26.0	23.4	22.0	13.5	10.2		
		MCWBR	11.1	11.3	11.8	14.2	12.3	11.4	10.9	11.4	11.8	13.3	11.0	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.324	0.381	0.418	0.406	0.403	0.410	0.425	0.375	0.355	0.331	0.297		
	taud		2.085	2.018	1.964	2.100	2.231	2.302	2.292	2.221	2.320	2.287	2.188	2.089		
	Ebn at Noon		219	245	252	259	269	271	266	255	256	236	204	195		
	Edh at Noon		25	36	46	45	42	39	39	40	32	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		426	795	1228	1509	1877	2073	1960	1683	1213	670	309	272		
	RadStd		43	44	132	146	161	136	105	124	98	108	36	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.76	N/A	N/A	N/A	N/A	+189	+151

Nomenclature: See separate page